

Supplementary Information for Risk-Aware Relational Spectral Recovery under Partial Quantum Observability

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S1 Global constraint and clock-conditioned dynamics

This note collects the background relational formulation used only in compressed form in the main Methods. The total Hilbert space is written as

$$\mathcal{H}_{\text{tot}} = \mathcal{H}_C \otimes \mathcal{H}_S, \quad (\text{S1.1})$$

and physical states are taken to satisfy the global constraint

$$\hat{H}_{\text{tot}}|\Psi\rangle = 0. \quad (\text{S1.2})$$

For a separable Hamiltonian

$$\hat{H}_{\text{tot}} = \hat{H}_C \otimes I_S + I_C \otimes \hat{H}_S, \quad (\text{S1.3})$$

a non-trivial product-basis solution requires matching clock and system energies,

$$\text{Spec}(\hat{H}_C) \cap \text{Spec}(-\hat{H}_S) \neq \emptyset. \quad (\text{S1.4})$$

Indeed, in the product eigenbasis of $\hat{H}_C$ and $\hat{H}_S$, each component has total energy $E_k^C + E_j^S$. A non-zero vector in the kernel can therefore have support only on pairs satisfying $E_k^C = -E_j^S$. This

is the finite-dimensional version of the compatibility condition used in the simulations. It is not used in the main text as an additional experimental assumption; it records why the clock-system construction is internally consistent.

This condition is used as a consistency requirement for constructing finite clock-system simulations; the main paper does not rely on a stronger claim that the finite simulation realizes a complete theory of quantum time.

Clock labels are generated by the clock orbit

$$|\tau\rangle_C = e^{-i\hat{H}_C\tau}|0\rangle_C. \quad (\text{S1.5})$$

Conditioning the global state on $|\tau\rangle_C$ gives

$$\rho_S(\tau) = \frac{\text{Tr}_C[(|\tau\rangle\langle\tau|_C \otimes I_S)|\Psi\rangle\langle\Psi|]}{\text{Tr}[(|\tau\rangle\langle\tau|_C \otimes I_S)|\Psi\rangle\langle\Psi|]}. \quad (\text{S1.6})$$

All trajectories in the manuscript are ordered families over these internal clock labels. They should not be read as externally time-parametrized histories. The finite clock representation also imposes resolution limits. Neighboring clock labels must be distinguishable enough to support separate conditional slices, but the clock orbit remains finite and can recur. These effects are handled by using a fixed ordered set of clock labels and by calibrating clock-consistency thresholds from reference trajectories rather than asserting an ideal continuum clock.

The implementation used for all reported policy comparisons fixes the clock object rather than tuning it per regime. The clock Hilbert space is one qubit, the initial clock state is $|+\rangle_C$, and

$$H_C = 0.8X_C. \quad (\text{S1.7})$$

The sampled internal labels are

$$\tau_r \in \{-2.0, -1.5, -1.0, -0.5, 0, 0.5, 1.0, 1.5, 2.0\}, \quad m = 9. \quad (\text{S1.8})$$

The same clock Hamiltonian, clock grid, and ordering convention are used for the syndrome-first candidate, the relational candidate, the reference bank, and all policy comparisons.

S2 Spectral deformation and detectability

At each clock slice, the system state $\rho_S(\tau)$ is mapped to a weighted relational graph with adjacency matrix $W^{(\tau)}$. The graph Laplacian is

$$L(\tau) = D^{(\tau)} - W^{(\tau)}, \quad D_{ii}^{(\tau)} = \sum_{j \neq i} W_{ij}^{(\tau)}. \quad (\text{S2.1})$$

The graph functional is fixed by the code family in the reported experiments. For the phase-flip repetition code, the edge weight is the absolute two-qubit coherence

$$W_{ij}^{(\tau)} = \sum_{a \neq b} \left| \left(\rho_{ij}^{(\tau)} \right)_{ab} \right|. \quad (\text{S2.2})$$

For the bit-flip repetition code, the edge weight is the two-qubit mutual information

$$W_{ij}^{(\tau)} = S(\rho_i^{(\tau)}) + S(\rho_j^{(\tau)}) - S(\rho_{ij}^{(\tau)}), \quad (\text{S2.3})$$

where $S(\rho) = -\text{Tr}(\rho \log \rho)$. Other relational functionals are available in the codebase for diagnostics, but they are not mixed into the reported C_2 versus C_3^{R} comparisons. The code-dependent choice is fixed before policy evaluation and is not selected per regime or per noise family. We use the Laplacian spectrum rather than the density-matrix spectrum directly because the decision variable is the collective connectivity pattern induced by code-specific two-qubit relations, not the global eigenvalue distribution of the state itself. Under a node relabeling the Laplacian changes only by permutation similarity, so its spectrum summarizes relational structure without depending on the arbitrary ordering of the three data qubits. Noise and partial observation induce perturbations

$$L(\tau) \mapsto L(\tau) + \Delta L(\tau). \quad (\text{S2.4})$$

If $\lambda_k(\tau)$ and $\tilde{\lambda}_k(\tau)$ denote the ordered spectra before and after perturbation, Weyl-type perturbation control gives

$$|\tilde{\lambda}_k(\tau) - \lambda_k(\tau)| \leq \|\Delta L(\tau)\|_2. \quad (\text{S2.5})$$

This bound explains why the Laplacian spectrum is an appropriate diagnostic object: perturbations in the relational graph create controlled spectral deformations, and the trajectory-level distance accumulates those deformations over clock labels. The experiments do not require every microscopic error to be separately identifiable. They require only that the deformation relevant to the recovery decision be large enough relative to calibrated background variation.

Consequently, non-trivial spectral deformations are detectable in principle when they are resolved by the spectral estimator and exceed the calibrated background variation. In the numerical implementation, this principle is used conservatively: detection and recovery are judged by empirical trajectory distances, calibrated admissibility penalties, and lower-tail recovery metrics rather than by a claim of universal noise detectability.

S3 Relational trajectory-space formalism

The clock-conditioned spectra define a spectral family

$$\Gamma_\Psi(\tau) = \Lambda(\tau) = (\lambda_1(\tau), \dots, \lambda_n(\tau)). \quad (\text{S3.1})$$

For sampled clock labels $\{\tau_r\}_{r=1}^m$, the computational distance between two spectral families is

$$\begin{aligned} d_{\mathcal{M}}^{ab} &\equiv d_{\mathcal{M}}(\Gamma^{(a)}, \Gamma^{(b)}), \\ d_{\mathcal{M}}^{ab} &= \left(\sum_{r=1}^m w_r \left\| \Lambda^{(a)}(\tau_r) - \Lambda^{(b)}(\tau_r) \right\|_2^2 \right)^{1/2}. \end{aligned} \quad (\text{S3.2})$$

We use the term relational spectral manifold only as a geometric shorthand for this constrained metric trajectory space. The reported experiments use the sampled spectra and the metric $d_{\mathcal{M}}$ directly. Differential-geometric terms such as tangent vectors, geodesic curves, and Christoffel symbols are therefore interpretive language, not assumptions required by the recovery algorithm or by the policy validation. Boundary points, discrete reference banks, and piecewise admissible regions are allowed in the computational formulation.

This is also why the construction is not an ordinary temporal smoother. A standard smoother regularizes an externally ordered sequence. Here the ordering comes from an internal clock-conditioned family, and the same candidate must satisfy Laplacian validity, clock consistency, spectral smoothness, and reference-anchor compatibility. The role of this section is therefore claim calibration: it explains the metric object being audited without asserting a foundational smooth-manifold theory.

S4 Admissibility penalties and recovery algorithm

The admissible set is represented as an intersection of constraint families,

$$\mathcal{A} = \mathcal{A}_{\text{lap}} \cap \mathcal{A}_{\text{rel}} \cap \mathcal{A}_{\text{ref}}. \quad (\text{S4.1})$$

The implementation audits membership through four penalties. The Laplacian penalty checks whether each slice remains a valid weighted-graph Laplacian:

$$\begin{aligned} \Phi_{\text{lap}}(\Gamma) = \sum_{r=1}^m & \left(\|L_r - L_r^\top\|_F^2 + \|L_r \mathbf{1}\|_2^2 + \sum_k [-\lambda_k(L_r)]_+^2 \right. \\ & \left. + \sum_{i \neq j} [L_{r,ij}]_+^2 + \left\| \text{diag}(L_r) + \sum_{j \neq i} L_{r,ij} \right\|_2^2 \right), \end{aligned} \quad (\text{S4.2})$$

where $[x]_+ = \max(x, 0)$. The terms enforce symmetry, zero row sum, positive-semidefinite spectrum, non-positive off-diagonal entries, and diagonal consistency.

The smoothness penalty measures slice-to-slice spectral jumps:

$$\Phi_{\text{smooth}}(\Gamma) = \sum_{r=1}^{m-1} \frac{\|\Lambda(\tau_{r+1}) - \Lambda(\tau_r)\|_2^2}{(\tau_{r+1} - \tau_r)^2}. \quad (\text{S4.3})$$

The clock-consistency penalty makes a large spectral jump more severe when the neighboring clock states are not well separated:

$$\Phi_{\text{clock}}(\Gamma) = \sum_{r=1}^{m-1} \frac{\|\Lambda(\tau_{r+1}) - \Lambda(\tau_r)\|_2}{1 - |\langle \tau_r | \tau_{r+1} \rangle|^2 + \epsilon}. \quad (\text{S4.4})$$

The reference-bank penalty is

$$\Phi_{\text{ref}}(\Gamma) = \min_{\Gamma' \in \mathcal{R}} d_{\mathcal{M}}(\Gamma, \Gamma')^2. \quad (\text{S4.5})$$

Thresholds are calibrated from reference trajectories as

$$\theta_k = \mu_k + \kappa \sigma_k, \quad \kappa = 1.5, \quad (\text{S4.6})$$

with leave-one-out evaluation for the reference penalty so that a reference trajectory does not calibrate its own distance to zero. For a candidate Γ_X , the normalized violation used by the robust policy is

$$v_X = \max_k \max \left\{ 0, \frac{\Phi_k(\Gamma_X) - \theta_k}{\theta_k + \epsilon} \right\}. \quad (\text{S4.7})$$

The use of a maximum rather than an average is important for the robust policy. Average violation can hide a single severe constraint failure, while the maximum records the worst active excursion beyond threshold. This is the value denoted by v_A and v_B in the C_3^R gate.

The reference bank contains three clean trajectories for each encoded code: the logical reference state and two $\pm 20^\circ$ logical perturbations. These reference trajectories use the same clock grid, clock Hamiltonian, and graph functional as the observed row, but the noise channel is removed. The reference anchor is selected only by trajectory distance:

$$\Gamma_{\text{ref,anchor}} = \arg \min_{\Gamma' \in \mathcal{R}} d_{\mathcal{M}}(\Gamma', \Gamma_{\text{obs}})^2. \quad (\text{S4.8})$$

No clean logical outcome, candidate fidelity, or oracle row label is used during anchor selection. This separation is important because the reference anchor is part of candidate construction, whereas fidelity and oracle quantities are held out for evaluation. The three-anchor bank is intentionally minimal. It is used to test whether a small auditable reference family can support recovery-side control, not to approximate a dense clean-state manifold. The recovery objective is

$$J_{\text{rec}}(\Gamma) = \alpha d_{\mathcal{M}}(\Gamma, \Gamma_{\text{obs}})^2 + \beta d_{\mathcal{M}}(\Gamma, \Gamma_{\text{ref,anchor}})^2 + \lambda_1 \Phi_{\text{lap}}(\Gamma) + \lambda_2 \Phi_{\text{smooth}}(\Gamma) + \lambda_3 \Phi_{\text{clock}}(\Gamma). \quad (\text{S4.9})$$

The main experiments use the discrete admissible projection as the default operator. A continuous convex refinement over local admissible combinations can be used as a diagnostic extension, but it is not required for the main reported policy results. The reason is operational: the first-stage operator selects from a finite, predeclared candidate set, so each accepted recovery can be traced to a reference anchor, a recovery objective value, and a vector of admissibility violations. This keeps the candidate construction separate from the downstream fidelity evaluation and prevents the reported policy result from depending on hidden continuous optimization flexibility. The discrete projection stage evaluates a finite set of reference-compatible candidate trajectories and selects the one minimizing the recovery objective while respecting admissibility. If $\mathcal{C}_{\text{ref}}$ is the finite reference-compatible candidate set, the first-stage recovery is

$$\Gamma_1 = \arg \min_{\Gamma \in \mathcal{C}_{\text{ref}}} J_{\text{rec}}(\Gamma) \quad \text{s.t.} \quad \Phi_k(\Gamma) \leq \theta_k \quad \forall k. \quad (\text{S4.10})$$

The optional refinement stage considers convex combinations of nearby admissible candidates. For a local neighborhood $\mathcal{N}_K(\Gamma_1) = \{\Gamma_j\}_{j=1}^K$, the feasible hull is

$$\mathcal{H}_K(\Gamma_1) = \left\{ \sum_{j=1}^K \alpha_j \Gamma_j : \alpha_j \geq 0, \sum_{j=1}^K \alpha_j = 1 \right\}. \quad (\text{S4.11})$$

The diagnostic second-stage candidate is then

$$\Gamma_2 = \arg \min_{\Gamma \in \mathcal{H}_K(\Gamma_1)} J_{\text{rec}}(\Gamma) \quad \text{s.t.} \quad \Phi_k(\Gamma) \leq \theta_k \quad \forall k. \quad (\text{S4.12})$$

The reported C3R policy comparisons use the conservative application rule

$$\Gamma_{\text{rec}} = \Gamma_1, \quad (\text{S4.13})$$

while Γ_2 is retained only for sensitivity diagnostics. This two-stage design keeps the paper-facing operator auditable while leaving room for continuous diagnostics. In particular, it avoids post-hoc candidate shaping, preserves row-level interpretability, and makes the intervention metrics attributable to the policy gate rather than to an additional refinement step. Stage 2 is therefore reported as a diagnostic stress test of the recovery geometry, not as a second source of final-policy performance.

S5 Encoded-QEC simulation protocol

The encoded-QEC testbed uses three-qubit repetition codes in bit-flip and phase-flip configurations. Noise families include bit-flip, phase-flip, depolarizing, dephasing, and mixed-Pauli channels. Each simulation row constructs two candidates:

$$A = \text{syndrome-first decoded recovery}, \quad B = \text{relational admissible recovery}. \quad (\text{S5.1})$$

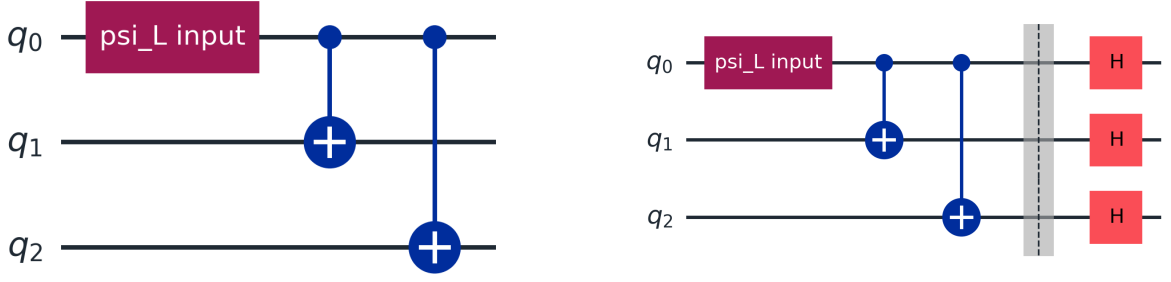


Figure S1: Encoder schematics for the encoded-QEC testbed. The left panel shows the bit-flip repetition-code encoder and the right panel shows the phase-flip encoder obtained by adding Hadamards on all three data qubits. The label ψ_L denotes the logical input state used to initialize the repetition-code block.

Candidate *A* is the conventional decoder-facing baseline. Candidate *B* is the output of admissible spectral recovery. The policy layer never changes the candidate construction; it only decides which candidate to trust on each row. This design isolates policy behavior. If the candidates themselves changed across partial, noisy, or ambiguous syndrome protocols, it would be difficult to tell whether an observed improvement came from better recovery construction or from a better decision rule. Here the decision layer is tested against a fixed pair of candidate recoveries, while syndrome observability affects the policy evidence and uncertainty diagnostics.

Syndrome observability is controlled by an observation ratio, syndrome-bit noise, ambiguity, measurement error, reset error, and measured information loss. The uncertainty proxy used by the robust gate is

$$u_{\text{raw}} = (1 - r_{\text{obs}}) + p_{\text{noise}} + a_{\text{amb}} + p_{\text{meas}} + p_{\text{reset}} + \ell_{\text{syn}}, \quad u_{\text{syn}} = \text{clip}_{[0,1]}(u_{\text{raw}}). \quad (\text{S5.2})$$

Figures S1 and S2 show the explicit circuit-level implementation of the testbed. The encoder circuits draw the three-qubit repetition-code conventions: bit-flip protection uses two CNOTs from the data qubit q_0 to q_1 and q_2 , while phase-flip protection appends Hadamards on all three qubits to map the bit-flip stabilizers into the X basis. The recovery circuits draw the corresponding syndrome-extraction circuits with two ancillas a_0, a_1 measured into classical bits s_0, s_1 and feed-forward X corrections gated on the syndrome values $\{0x1, 0x2, 0x3\}$. The clock-conditioned data state $\rho_{\text{data}}(\tau)$ is injected on $\{q_0, q_1, q_2\}$ at each clock slice; the relational graph functional above is computed from this injected state before any syndrome operation. The stress protocols used in the main analysis are clean syndrome, partial syndrome, noisy syndrome, partial-plus-noisy syndrome, and ambiguity with measurement/reset errors. The clean map identifies operating zones when direct syndrome information is reliable. The partial and partial-plus-noisy maps test missing evidence and combined degradation. The noisy-only map tests random syndrome-bit corruption without reducing observation ratio. The ambiguity-plus-measurement map tests whether uncertainty due to ambiguous or physically corrupted syndrome evidence produces the same risk-control behavior.

S6 Policy maps and operating-zone analysis

The four policies compared in the manuscript are deterministic maps from the same two candidates to one selected recovery. C_1 is the conservative rule-based hybrid. C_2 is the balanced proposal policy

with score

$$S_X^{(2)} = \lambda_s s_X - \lambda_t d_X - \lambda_i(1 - a_X) + \lambda_o \frac{1}{1 + J_X}. \quad (\text{S6.1})$$

C_3 is a hard-admissibility ablation: it selects the uniquely admissible candidate when exactly one candidate is admissible, and otherwise uses a safety-weighted score. The clean-regime result that C_3 does not form an independent operating zone is interpreted as an ablation result, not as the central claim of the paper. The reason for retaining C_3 is methodological. It tests the simpler hypothesis that strict admissibility alone is enough to create a useful safety-first region. The negative operating-zone result indicates that this Boolean criterion is too coarse once inadmissibility has already been priced into the balanced score used by C_2 . This motivates the one-sided veto design of C_3^R .

The revised policy C_3^R treats C_2 as a proposal and applies a one-sided veto:

$$C_3^R = B \iff [C_2 = B] \wedge [\Delta S \geq \delta_S] \wedge [v_A \geq \delta_A] \wedge [B \text{ admissible}, v_B \leq \delta_B] \wedge [u_{\text{syn}} \leq u_{\text{max}}]. \quad (\text{S6.2})$$

The main operating point is

$$\delta_S = 0, \quad \delta_A = 0, \quad \delta_B = 0, \quad u_{\text{max}} = 0.99. \quad (\text{S6.3})$$

Thus

$$C_3^R = B \Rightarrow C_2 = B, \quad (\text{S6.4})$$

so the revised controller can block proposed relational switches but cannot create new ones. The useful diagnostics for C_3^R are therefore conditional on C_2 proposing B . Unconditional switch rates can be misleading because most rows may already be preserved by C_2 . The intervention analysis in the main text and result tables consequently reports block rate given $C_2 = B$, harmful-block precision, harmful-switch recall, beneficial-switch retention, and net intervention gain.

Table S1: Clean-regime preferred-policy counts over the 240-cell operating map. Each cell is averaged over ten random seeds.

Policy	Cells	Share	Interpretation
C_1	197	0.8208	Default conservative hybrid region.
C_2	43	0.1792	Structured correction region; all cells are phase-flip code rows.
C_3	0	0.0000	No independent strict-admissibility zone.
C_3^R	0	0.0000	No clean-regime veto trigger at the default gate.

Table S2: Noise-family composition of the clean C_2 preferred region.

Noise family	C_2 -preferred cells
Bit flip	2
Dephasing	16
Depolarizing	6
Mixed Pauli	9
Phase flip	10

Table S3: Partial-syndrome preferred-policy counts by observed syndrome ratio.

r_{obs}	Cells	C_1	C_2	C_3	C_3^{R}
1.00	72	37	35	0	0
0.75	72	72	0	0	0
0.50	72	12	0	0	60
0.25	72	14	0	0	58

Table S4: Noisy-only preferred-policy counts by syndrome-bit noise probability. The robust policy remains inactive in this negative-control map.

p_{noise}	Cells	C_1	C_2	C_3	C_3^{R}
0.00	72	37	35	0	0
0.01	72	49	23	0	0
0.03	72	51	21	0	0
0.05	72	51	21	0	0
0.10	72	61	11	0	0

Table S5: Representative partial-plus-noisy policy-map rows showing regime cells where C_3^{R} is the preferred policy. Each row contains 72 cells.

r_{obs}	p_{noise}	C_1	C_2	C_3	C_3^{R}
0.50	0.03	13	0	0	59
0.50	0.05	12	0	0	60
0.50	0.10	15	0	0	57

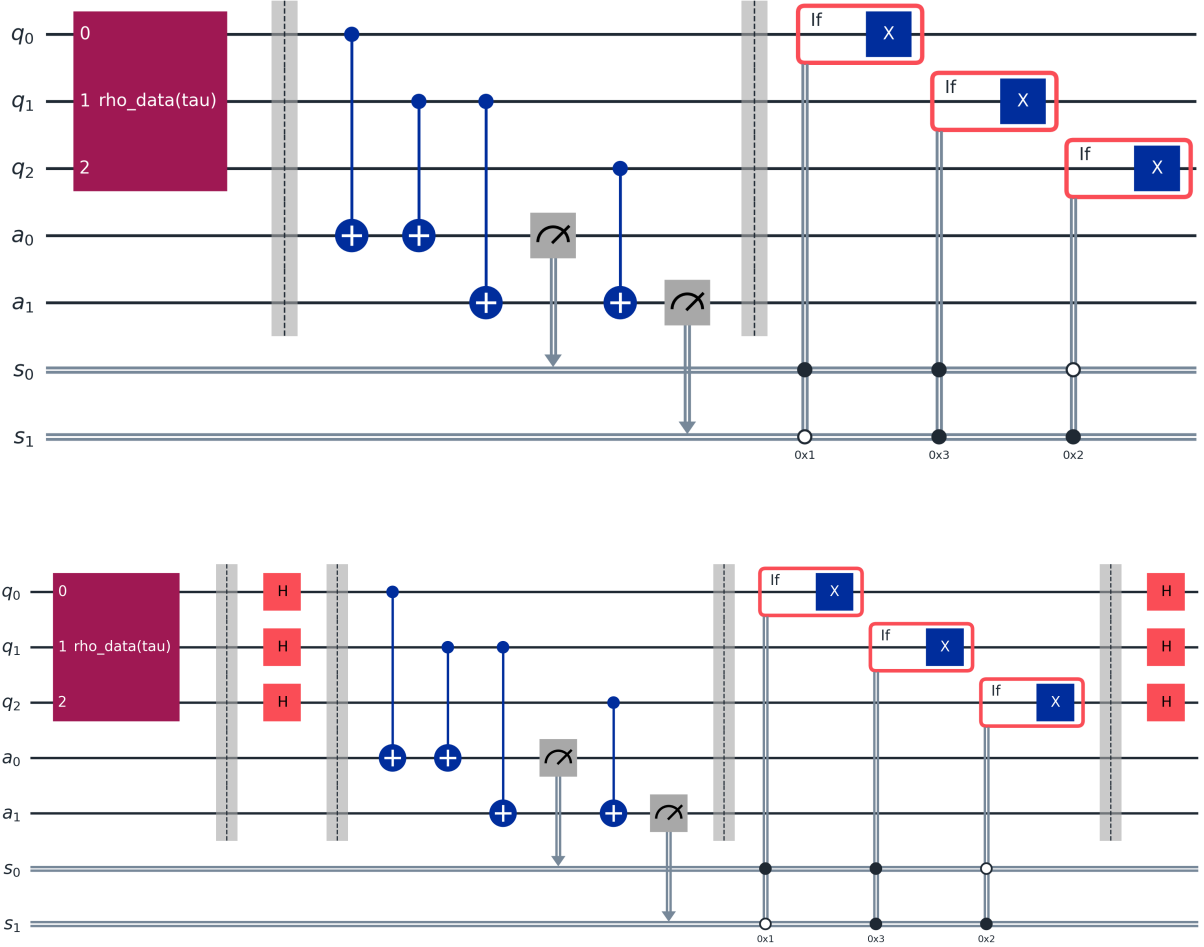


Figure S2: Syndrome-extraction and feed-forward recovery schematics. The upper panel shows the bit-flip recovery circuit and the lower panel shows the phase-flip recovery circuit. The state $\rho_{\text{data}}(\tau)$ denotes the clock-conditioned three-qubit data-register state injected before syndrome extraction. Ancillas a_0, a_1 store the two parity checks and are measured into the classical register s_0, s_1 , with s_0 the least significant bit. The feed-forward integer convention is therefore $s_1 s_0$: 0x1 = 01 applies X to q_0 , 0x3 = 11 applies X to q_1 , and 0x2 = 10 applies X to q_2 . For the phase-flip circuit, Hadamards before and after syndrome extraction move the same correction logic into the phase-flip protection basis. The simulations trace out the ancilla register after recovery; the schematic is drawn in system-qubit order q_0, q_1, q_2 .

S7 Metric definitions

Let ΔF_C be the logical-fidelity gain of policy C . The false-safe and success thresholds are fixed before policy comparison:

$$\varepsilon_F = 0.01, \quad F_{\text{success}} = 0.99. \quad (\text{S7.1})$$

Central performance is summarized by mean gain, logical-success rate

$$P[F_C \geq F_{\text{success}}], \quad (\text{S7.2})$$

and non-worsening rate

$$P[\Delta F_C \geq 0]. \quad (\text{S7.3})$$

Downside risk is summarized by false-safe fidelity rate,

$$\text{FS}_F(C) = P[\Delta F_C < -\varepsilon_F], \quad (\text{S7.4})$$

the lower quantile $q_{0.05}(C)$, and

$$\text{CVaR}_{0.05}(C) = \mathbb{E}_{\text{emp}}[\Delta F_C \mid \Delta F_C \leq q_{0.05}(C)]. \quad (\text{S7.5})$$

For C_3^{R} , the intervention metrics are conditional on C_2 proposing B . The block indicator is

$$B_R = \mathbf{1}[C_2 = B] \mathbf{1}[C_3^{\text{R}} = A]. \quad (\text{S7.6})$$

Harmful and beneficial proposed switches are

$$H_2 = \mathbf{1}[C_2 = B] \mathbf{1}[\Delta F_A > \Delta F_B + \varepsilon_F], \quad (\text{S7.7})$$

$$G_2 = \mathbf{1}[C_2 = B] \mathbf{1}[\Delta F_B > \Delta F_A + \varepsilon_F]. \quad (\text{S7.8})$$

We report block rate, harmful-block precision, harmful-switch recall, beneficial-switch retention, prevented loss, missed gain, and net intervention gain. The uncertainty-bin analysis groups rows by u_{raw} to check whether the robust policy behaves as an uncertainty-conditioned intervention rather than as an unconditional score adjustment. The scalar net intervention effect is

$$E_{\text{net}} = \sum_i B_{R,i} (\Delta F_{A,i} - \Delta F_{B,i}), \quad (\text{S7.9})$$

equivalently,

$$E_{\text{net}} = \sum_i B_{R,i} \max\{0, \Delta F_{A,i} - \Delta F_{B,i}\} - \sum_i B_{R,i} \max\{0, \Delta F_{B,i} - \Delta F_{A,i}\}. \quad (\text{S7.10})$$

Thus the same scalar can be read as prevented harmful-switch loss minus missed beneficial-switch gain. A positive value means that blocked switches improved aggregate fidelity relative to accepting the C_2 proposal. We also report $E_{\text{net}} / \sum_i B_{R,i}$ and $E_{\text{net}} / \sum_i \mathbf{1}[C_{2,i} = B]$ to separate aggregate benefit from the number of vetoed proposals. For offline comparison we also compute oracle regret,

$$\Delta F^* = \max\{\Delta F_A, \Delta F_B\}, \quad \text{Reg}(C) = \Delta F^* - \Delta F_C. \quad (\text{S7.11})$$

The oracle is not available to the policy; it is used only to quantify how far the chosen candidate is from the best candidate available on that simulation row. This diagnostic is especially useful when mean gain and false-safe rate point in different directions.

S8 Threshold and uncertainty diagnostics

This section audits the two threshold choices that could otherwise appear arbitrary: the uncertainty gate $u_{\max}$ in the robust policy and the admissibility calibration factor κ . The purpose is not to tune an optimum threshold after seeing the outcomes, but to show how the reported operating point behaves relative to nearby counterfactual settings and negative controls.

The default robust gate uses $u_{\max} = 0.99$. This value should not be interpreted as a calibrated posterior confidence level. It is a saturation guard for the additive stress index u_{raw} after clipping. Table S6 reports offline counterfactuals for the same rows under alternative $u_{\max}$ values.

Table S6: C_3^R uncertainty-threshold sensitivity. Mean gain and false-safe fidelity rate are recomputed from the same result rows after changing only the $u_{\max}$ gate. Noisy-only is included as a negative control; $u_{\max}$ has no effect there because no row triggers the robust gate.

Regime	$u_{\max}$	mean gain	FS_F	blocked	precision	ben. block	net/block
Noisy-only	any	0.1169	0.0678	0	–	–	–
Partial	0.75	0.1189	0.0340	543	0.8508	0.4833	0.0106
Partial	0.99	0.1189	0.0389	476	0.8697	0.3750	0.0119
Partial	1.00	0.1169	0.0694	0	–	–	–
Partial+noisy	0.75	0.1205	0.0000	821	0.8465	1.0000	0.0092
Partial+noisy	0.99	0.1212	0.0134	670	0.8791	0.6739	0.0136
Partial+noisy	1.00	0.1170	0.0685	0	–	–	–
Ambig.+meas.	0.75	0.1608	0.0000	480	0.7896	0.5316	0.0061
Ambig.+meas.	0.99	0.1604	0.0000	357	0.7759	0.4211	0.0044
Ambig.+meas.	1.00	0.1600	0.0000	0	–	–	–

The admissibility thresholds also contain the calibration constant κ in $\theta_k = \mu_k + \kappa\sigma_k$. To check that the default $\kappa = 1.5$ is not a single-point artifact, we ran the recovery-ablation sweep with $\kappa \in \{1.0, 1.5, 2.0\}$ while holding the noise families, noise strengths, and reference-bank widths fixed. The sweep covers bit-flip, phase-flip, and dephasing stress scenarios at five strengths and three bank widths, giving 45 cases per calibration point.

Table S7: Admissibility-calibration sensitivity for the recovery objective. Recovered admissible is the calibrated-admissibility rate of the final recovered trajectory. Candidate improvement is the rate at which the optional Stage-2 admissible candidate improves the recovery objective relative to Stage 1, and helpful tradeoff additionally requires non-worse clean distance.

κ	cases	rec. admiss.	cand. improve	mean cand. ΔJ	helpful
1.0	45	0.2000	0.5556	-0.1100	0.1111
1.5	45	1.0000	0.6667	0.1702	0.1111
2.0	45	1.0000	0.6667	0.1702	0.1111

The default $\kappa = 1.5$ is on the same plateau as $\kappa = 2.0$ in this sweep, whereas the stricter $\kappa = 1.0$ calibration sharply reduces recovered admissibility and gives a negative mean candidate objective gain. The helpful tradeoff rate is invariant across κ in this sweep, indicating that these κ changes affect admissibility coverage and Stage-2 candidate acceptance more than the underlying clean-distance behavior. This supports using $\kappa = 1.5$ as a moderate nondegenerate operating point, but it should not be interpreted as a universally optimized threshold. No Stage-2 refinements were accepted by the conservative application rule in this sweep, so Table S7 is calibration evidence for the admissibility side of the method rather than a separate final-policy performance claim.

These diagnostics show that the default $u_{\max} = 0.99$ gate behaves primarily as a saturation guard. When $u_{\max} = 1.00$, the veto is disabled in the reported degraded regimes and C_3^R collapses

Table S8: Uncertainty-bin intervention summary at the default $u_{\max} = 0.99$. Bins with zero blocks show that the gate is concentrated near saturated syndrome stress rather than acting as an unconditional score penalty.

Regime	u_{raw} bin	cases	$C_2 = B$	blocked	block rate	precision	net
Partial	< 1.00	1638	618	0	0.0000	–	0.0000
Partial	$[1, \infty)$	1242	476	476	1.0000	0.8697	5.6616
Partial+noisy	$[0.75, 1.00)$	516	180	29	0.1611	0.6897	-0.3097
Partial+noisy	$[1, \infty)$	1644	641	641	1.0000	0.8877	9.4309
Ambig.+meas.	< 1.00	2142	538	0	0.0000	–	0.0000
Ambig.+meas.	$[1, \infty)$	1458	357	357	1.0000	0.7759	1.5749

to C_2 . When $u_{\max} = 0.75$, the controller becomes more conservative, reducing false-safe failures further in some regimes but increasing beneficial-switch loss. The negative net value in the partial-plus-noisy $[0.75, 1.00)$ bin is a small borderline-stress region where the default gate vetoes some beneficial C_2 switches; the aggregate partial-plus-noisy benefit is dominated by the saturated-stress bin $[1, \infty)$. In the $u_{\max} = 0.75$ counterfactual for the same regime, the aggressive threshold blocks all beneficial C_2 switches, eliminating the upside from the relational candidate. Therefore, the default operating point should be interpreted as a safety-biased but not optimized threshold.

S9 Supplementary result summaries

This section reports the numerical values underlying the main Results. The main text emphasizes operating-zone interpretation and downside-risk behavior; the tables below retain additional central-performance, oracle, and intervention diagnostics.

Table S9: Full regime-level comparison between C_2 and C_3^R . Entries are reported as C_2/C_3^R .

Regime	n	mean gain	logical success	non-worsen	FS_F	true failure boundary
Clean	2400	0.1273/0.1273	0.6358/0.6358	0.9554/0.9554	0.0238/0.0238	0.2571/0.2571
Partial	2880	0.1169/0.1189	0.4552/0.5976	0.8774/0.9306	0.0694/0.0389	0.3611/0.2653
Noisy-only	3600	0.1169/0.1169	0.4556/0.4556	0.8814/0.8814	0.0678/0.0678	0.3589/0.3589
Partial+noisy	2160	0.1170/0.1212	0.4551/0.7241	0.8792/0.9755	0.0685/0.0134	0.3616/0.1819
Ambig.+meas.	3600	0.1600/0.1604	0.5208/0.5978	1.0000/1.0000	0.0000/0.0000	0.2136/0.1847

Table S10: Lower-tail and oracle-regret diagnostics. Entries are reported as C_2/C_3^R .

Regime	$q_{0.05}$	CVaR $_{0.05}$	mean oracle regret	raw uncertainty mean
Clean	0.0000/0.0000	-0.0033/-0.0033	0.0132/0.0132	0.0000
Partial	-0.0152/-0.0006	-0.0208/-0.0147	0.0271/0.0251	0.7604
Noisy-only	-0.0104/-0.0104	-0.0207/-0.0207	0.0270/0.0270	0.0380
Partial+noisy	-0.0104/0.0000	-0.0206/-0.0034	0.0271/0.0228	1.0989
Ambig.+meas.	0.0192/0.0238	0.0078/0.0084	0.0336/0.0332	0.8261

The intervention table should be read together with the negative controls. In the clean and noisy-only regimes, C_3^R blocked no C_2 switches. Thus the non-zero intervention rows are not evidence of a global conservative bias; they identify the regimes where the uncertainty and admissibility gates made the C_2 proposal unsafe enough to veto.

Table S11: C_3^R intervention quality in regimes with non-zero blocking.

Regime	$C_2 = B$	blocked	block rate	precision	recall	ben. block	net effect
Partial	1094	476	0.4351	0.8697	0.4461	0.3750	5.6616
Partial+noisy	821	670	0.8161	0.8791	0.8475	0.6739	9.1212
Ambig.+meas.	895	357	0.3989	0.7759	0.3929	0.4211	1.5749

Table S12: Normalized intervention effects. Prevented and missed values are aggregate fidelity-gain differences over blocked C_2 switches.

Regime	blocked	prevented	missed	net	net/block	net/($C_2 = B$)
Partial	476	12.2670	6.6055	5.6616	0.0119	0.0052
Partial+noisy	670	17.4350	8.3139	9.1212	0.0136	0.0111
Ambig.+meas.	357	8.3526	6.7777	1.5749	0.0044	0.0018